Activity - 5

Procedure

1. Draw a circle with centre O and with suitable radius.
2. Make it a semi-circle through folding. Consider the point A, B on it.
3. Make crease along AB in the semi circles and open it.
4. We get one more crease line on the another part of semi circle, name it as CD ($observe\ AB = CD$)
5. Join the radius to get the ΔOAB and ΔOCD.
6. Using trace paper, take the replicas of triangle ΔOAB and ΔOCD.
7. Place these triangles ΔOAB and ΔOCD one on the other.

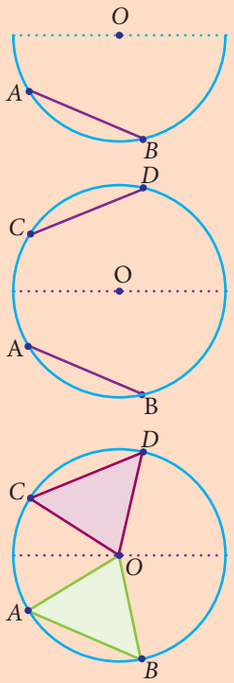


Fig. 4.60

Observation

1. What do you observe? Is $\Delta OAB \equiv \Delta OCD$?
2. Construct perpendicular line to the chords AB and CD passing through the centre O. Measure the distance from O to the chords.

Now we are going to find out the length of the chords AB and CD, given the angles subtended by two chords at the centre of the circle are equal. That is, $\angle AOB = \angle COD$ and the two sides which include these angles of the ΔAOB and ΔCOD are radii and are equal.

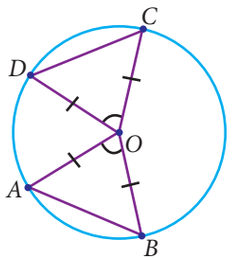


Fig. 4.61

By SAS rule, $\Delta AOB \equiv \Delta COD$. This gives chord AB = chord CD. Now let us write the converse result as follows:

Converse of theorem 8

If the angles subtended by two chords at the centre of a circle are equal, then the chords are equal.

In the same way we are going to discuss about the distance from the centre, when the equal chords are given. Draw the perpendicular $OL \perp AB$ and $OM \perp CD$. From theorem 7, these perpendicular divides the chords equally. So $AL = CM$. By comparing the ΔOAL and ΔOCM, the angles $\angle OLA = \angle OMC = 90°$ and $OA = OC$ are radii. By RHS rule, the $\Delta OAL \equiv \Delta OCM$. It gives the distance from the centre $OL = OM$ and write the conclusion as follows.

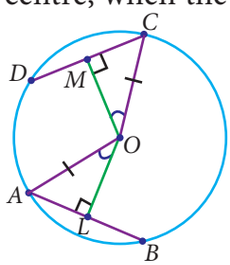


Fig. 4.62

Theorem 9 Equal chords of a circle are equidistant from the centre.

Let us know the converse of theorem 9, which is very useful in solving problems.

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